CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: Bazar Koro

Group No: 07, CSE471 Lab Section: 08, Spring 2026		
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Project Overview

"Bazar Koro" is a game-changing, hyper-local marketplace because it empowers neighborhood businesses to compete against giant "dark store" delivery apps without the burden of inventory costs. By uniquely integrating a "Marketer" role, the platform creates a self-sustaining, community driven advertising network where locals earn commissions for promoting nearby shops. This asset-light model has massive scaling potential; it doesn't require expensive warehouses, just a robust digital infrastructure to connect buyers, sellers, drivers, and marketers. It champions the local economy, offers diverse, niche products that big corporations ignore, and builds a resilient, community-focused ecosystem with virtually unlimited growth potential across different cities.

Tech Stack

- **Language:** JavaScript ,TypeScript
- **Framework:** React.js (Frontend) & Express.js running on Node.js (Backend)
- **Styling:** Tailwind CSS
- **Database:** MongoDB
- **ORM:** Mongoose/ MongoDB cluster
- **Deployment:** Hostinger
- **External APIs:**
    - Bkash(For handling buyer checkouts and marketer ad payments)
    - Cloudinary API (For storing seller product images)
    - Google Maps API (For delivery driver routing)
    - SendGrid API (For automated email marketing)

User Roles

1. **Buyer:** Browses local shops, adds items to their unified shopping cart, leaves reviews, and purchases products.
2. **Seller (Shop Owner):** A verified local business that manages their public storefront, uploads inventory, tracks sales analytics, and accepts/prepares incoming orders.
3. **Delivery Driver:** A freelance logistics partner who toggles active status, claims ready orders from shops, and updates the delivery status until it reaches the buyer's door.

4. **Marketer:** A community promoter who selects products to boost, sets ad budgets, schedules campaigns, and generates affiliate links to promote local items for a commission.
5. **Admin:** A system super-user responsible for verifying shop registrations, monitoring system activity, managing platform commissions, and resolving user disputes.

Functional Requirements

SL	Common Workflow
1	Registration and Login System: Users can create an account by providing basic details such as name, email, phone number, and password. The system uses Role-Based Access Control (RBAC), allowing a single user account to hold multiple roles (Buyer, Seller, Driver, Marketer). Once registered, users can log in securely and toggle between their dashboards.
2	Admin Role & Access Management: An authorized administrator signs in to a dedicated panel to review pending store applications, check submitted details (e.g., trade license, location), and approve or reject them. The admin also monitors platform revenue and resolves disputes.

Module 1

Member	Feature Description
1	Unified Role Switching Dashboard: A dynamic dashboard interface that detects the user's active roles. It allows a user to smoothly switch their UI view from "Buyer Mode" to "Seller Mode" or "Marketer Mode," loading the appropriate data and permissions instantly.
1	Dynamic Shopping Cart: A responsive cart system where buyers can add products from different local stores, update quantities, and view real-time subtotals. The cart visually groups items by store.
1	Search & Filter Engine: A robust search bar allowing buyers to find specific products or stores. It includes advanced filters so users can sort by categories (e.g., Groceries, Pharmacy), price range, or distance.
1	Payment Integration (External API): Secure checkout gateway using the Bkash API. Buyers enter payment details to pay for their carts. The system processes the payment, splits the funds (platform commission vs. seller payout), and generates a digital receipt.
1	Order History & "Buy Again": A dedicated profile section where buyers can view detailed invoices of past orders, check the live status of active orders, and use a 1-click "Buy Again" button to easily duplicate previous purchases.

1	Review and Rating System: After an order is marked "Delivered," buyers can leave a 1–5 star rating and a written review for both the product and the store, which displays on the seller's storefront.

Module 2

Member	Feature Description
2	Store Profile Creation: Sellers can create and customize their public storefront. They can input their business name, physical address, operating hours, and a short description.
2	Inventory Management: A comprehensive dashboard for sellers to easily add, edit, delete, and price their products. Sellers can organize their items into custom categories and toggle an item as "Out of Stock."
2	Order Management System: A real-time feed where sellers view incoming orders. They can accept or reject new orders, prepare the items, and change the status to "Ready for Pickup," which automatically alerts the delivery drivers.
2	Image Hosting (External API): Integration with the Cloudinary API to handle product image uploads. When a seller uploads a photo, the API automatically compresses, resizes, and stores the image securely.

2	Sales Analytics Dashboard: A visual reporting tab for sellers showing their weekly and monthly revenue, total completed orders, and automatically highlighting their "Top 3 Best-Selling" products.
2	Coupon & Discount Engine: Sellers can generate custom promotional codes (e.g., "LOCAL10"). The system validates these codes at checkout, ensures they haven't expired, and recalculates the buyer's cart total.

Module 3

Member	Feature Description
3	Driver Dashboard & Status: A dedicated portal for delivery drivers where they can toggle their availability status (Online/Offline). It also tracks their daily earnings, completed trips, and current active deliveries.
3	Order Bidding & Assignment: A live feed showing orders marked as "Ready for Pickup." Online drivers can view the pickup location, drop-off distance, and delivery fee, allowing them to claim the job with a single click.
3	Delivery Status Updates: Once an order is claimed, the driver controls the delivery timeline. They update the system status sequentially: "At Store," "Picked Up," "On the Way," and "Delivered."

3	Routing Integration (External API): Integration with the Google Maps API. When a driver accepts an order, the app generates an optimized, turn-by-turn route from their location, to the shop, and finally to the buyer.
3	Proof of Delivery System: To finalize a delivery, the driver must securely close the order by either uploading a photo of the delivered package at the door or entering a unique 4-digit PIN provided to the buyer.
3	Vehicle Profile & Filtering: Drivers specify their vehicle type in their profile (e.g., Bicycle, Motorcycle, Van). The system uses this data to filter order visibility based on order size and weight.

Module 4

Member	Feature Description
4	Marketer Dashboard & Budgeting: A specialized view for the "Marketer" role. Users select items they want to promote, set a monetary advertising budget, and pay for the campaign using the integrated payment system.
4	Promoted Listings Engine: The logic that modifies the buyer's search results and homepage. Products with active ad budgets are visually tagged as "Sponsored" and forced to appear at the very top of the feed.

4	Ad Analytics Tracking: A reporting tool for marketers to see the ROI of their ads. It tracks and displays metrics like "Total Impressions" and "Click-Through Rate."
4	Automated Newsletters (External API): Integration with the SendGrid API. The system compiles the top "Sponsored" products for the week and sends a formatted promotional email blast to registered buyers in that neighborhood.
4	Ad Campaign Scheduling: Marketers can dictate exactly when their ads run by setting specific "Start" and "End" dates and times for their campaigns.
4	Affiliate Link Generation: Marketers can generate unique, trackable URLs for any product. When they share this link externally and a user makes a purchase, the system tracks the conversion and credits the marketer.